

MODULE 41 · EXAM STRATEGY - ALL DOMAINS

Confusion Pairs

Thirty pairs, each divided by one property. Learn the divider, not the definitions.

DOMAIN Cross-cutting EXAM WEIGHT All four MODULE 41 of 44

For each pair, one property decides. **Learn the divider**, and the definitions stop mattering.

Governance and identity

PAIR	THE ONE PROPERTY THAT DIVIDES THEM
SCP vs RCP	SCP follows the identity; RCP follows the resource
Boundary vs session policy	Boundary is permanent on the identity; session policy is per-session
Preventive vs detective	Preventive refuses the call; detective reports afterwards
AD Connector vs Managed AD	Connector stores nothing and cannot be shared; Managed AD can hold trusts
User pool vs identity pool	User pool is who you are; identity pool grants AWS credentials
RAM vs VPC sharing	RAM shares a resource; VPC sharing shares subnets, owner keeps the plumbing

ON THE EXAM

The SCP and RCP divider is the one to over-learn. Cross-account access needs both sides to allow it, plus the KMS key.

Networking

PAIR	THE ONE PROPERTY THAT DIVIDES THEM
Peering vs Transit Gateway	Peering is non-transitive; TGW is a router with route tables
Association vs propagation	Association is which table I read; propagation is which tables learn about me
Inbound vs outbound Resolver	Inbound: on-premises asks AWS. Outbound: AWS asks on-premises
Gateway vs interface endpoint	Gateway is a route entry, VPC-only, free. Interface is an ENI address
Geolocation vs latency routing	Geolocation is a legal guarantee; latency is a performance preference
CloudFront vs Global Accelerator	CloudFront caches; Global Accelerator does not, and handles any protocol

ON THE EXAM

Association versus propagation is the most missed pair on the exam. Say it aloud until it is automatic.

Building

PAIR	THE ONE PROPERTY THAT DIVIDES THEM
Nested stacks vs cross-stack refs	Nested share one lifecycle; exports let stacks deploy independently
DeletionPolicy vs UpdateReplacePolicy	One covers stack deletion, the other covers replacement on update
Custom resource vs hook	A custom resource does work; a hook judges whether to proceed
Reserved vs provisioned concurrency	Reserved is capacity contention; provisioned is cold-start latency
Queue vs stream	A queue message is consumed and gone; a stream record stays
SNS vs EventBridge	SNS filters attributes; EventBridge routes on the event body

ON THE EXAM

Reserved concurrency is a ceiling and a floor at once. That dual nature is why one setting solves a noisy-neighbour problem.

Operating

PAIR	THE ONE PROPERTY THAT DIVIDES THEM
Governance vs compliance mode	Governance can be overridden by a privileged user; compliance cannot
GuardDuty vs Detective	GuardDuty detects; Detective explains a finding something else produced
Run Command vs State Manager	Run Command is now, once; State Manager is continuously, forever
Canaries vs RUM	Canaries run with no users; RUM needs real users
Control plane vs data plane	Control plane creates things; data plane uses what exists
Coverage vs utilisation	Low utilisation: bought too much. Low coverage: bought too little

ON THE EXAM

Coverage and utilisation point in opposite directions. Reading only one is how finance teams reach confident wrong conclusions.

Migrating

PAIR	THE ONE PROPERTY THAT DIVIDES THEM
MGN vs DRS	MGN migrates permanently; DRS protects production that stays put
Agent vs agentless discovery	Only the agent sees processes, which is what dependency mapping needs
Rehost vs relocate	Rehost changes where it runs; relocate changes only the hypervisor
Replatform vs refactor	Replatform changes a component; refactor changes the architecture
Homogeneous vs heterogeneous	Same engine means no schema conversion at all
Full load + CDC vs CDC-only	CDC-only needs a target already populated by something else

ON THE EXAM

MGN and DRS share most of their technology and answer opposite questions. Ask whether production is moving or staying.

How to use these under exam pressure

- ◆ **Name the pair the question is testing** before reading the options. It is usually one of these thirty.
- ◆ **Find the word that resolves it.** Never, guarantee, automatically, ongoing, unattended, or a number.
- ◆ **Two plausible options usually means one pair.** Two implausible ones are there to make it four.
- ◆ If both members of a pair still seem right, you have found the wrong pair. Read the requirement again.

ON THE EXAM

Most questions reduce to two credible options. The other two exist to fill the page, and eliminating them is fast.

A regulator requires that audit logs cannot be deleted by anyone, including the cloud team, and that the control cannot be removed by an administrator in the workload account. The account is a member of an organisation with the management account excluded from SCPs.

The logs are written to S3 in a dedicated Log Archive account.

Which combination is correct?

- ☐ A Object Lock in compliance mode, plus an RCP on the bucket so the control follows the resource
- ☐ B Object Lock in governance mode, plus an SCP on the workload account
- ☐ C Object Lock in compliance mode, plus an SCP on the management account
- ☐ D Versioning, plus a permissions boundary on administrators

◆ PAUSE AND DECIDE

Which combination is correct?

- A

 Compliance mode plus an RCP
Two pairs resolved: compliance cannot be overridden by anyone, and an RCP follows the resource so it applies regardless of which identity calls.
- B

 Governance plus SCP
Both halves wrong. Governance mode exists to be overridden, and an SCP follows the identity rather than the bucket.
- C

 Compliance plus SCP on management
Right on Object Lock, and SCPs do not apply to the management account, which the scenario states explicitly.
- D

 Versioning and a boundary
Versioning keeps old copies that a delete-capable administrator can also remove. A boundary caps an identity, not the resource.

THE SIGNAL

Two pairs in one question: governance versus compliance, and SCP versus RCP. Each half eliminates two options.

One Lambda function occasionally consumes most of the account's concurrency, throttling a payment handler. Separately, the payment handler's p99 latency spikes to 3 seconds on its first invocation after idle periods, and the business needs consistent latency.

Both problems must be solved.

What should the solutions architect implement?

- ☐ A Reserved concurrency on both functions, and provisioned concurrency on the payment handler
- ☐ B Provisioned concurrency on both functions
- ☐ C Reserved concurrency on the payment handler only
- ☐ D Request a concurrency quota increase, and enable SnapStart

◆ PAUSE AND DECIDE

What should the solutions architect implement?

- A

 Reserved on both, provisioned on the handler
Reserved caps the noisy function and guarantees the handler a share; provisioned keeps environments warm to remove the cold start. Two problems, two settings.
- B

 Provisioned on both
Solves latency twice and capacity contention never. The noisy function can still consume the account's concurrency.
- C

 Reserved on the handler only
Solves the contention and leaves the 3-second cold start. It answers one of two stated problems.
- D

 Quota increase and SnapStart
A bigger shared pool fails the same way, and SnapStart is Java, Python and .NET only, with no runtime stated.

THE SIGNAL

Two problems that sound like one. Capacity contention and cold-start latency need different settings with confusingly similar names.

On-premises servers cannot resolve the private DNS names of migrated services. Separately, a chatty application split across the two estates is now slow. The team proposes upgrading the Direct Connect link from 1 Gbps to 10 Gbps to fix both.

There is a Route 53 private hosted zone.

What should the solutions architect advise?

- ☐ A A Route 53 Resolver inbound endpoint for the DNS problem, and moving the chatty pair into one estate for the latency
- ☐ B Upgrade the link to 10 Gbps, which resolves both
- ☐ C A Route 53 Resolver outbound endpoint, and upgrade the link
- ☐ D Distribute hosts files, and increase the application's instance size

◆ PAUSE AND DECIDE

What should the solutions architect advise?

- A** Inbound endpoint, and stop the crossing
On-premises querying AWS is inbound. And latency is distance, so only removing the crossing fixes the chatty pair.
- B** Upgrade the link
Bandwidth is not latency, and no link speed makes AWS private zones resolvable from on-premises. It fixes neither problem.
- C** Outbound endpoint plus upgrade
Outbound is AWS querying on-premises, which is the opposite direction. And the upgrade still does not address latency.
- D** Hosts files and bigger instances
Bypasses DNS entirely, and the application is waiting on the network rather than short of CPU.

THE SIGNAL

Two pairs again: inbound versus outbound Resolver, and bandwidth versus latency. Both halves of option B are wrong.

A security team receives many related GuardDuty findings over a day but cannot see them as one incident. Once an incident is confirmed, they also need to determine which other resources the compromised role touched over the previous month.

GuardDuty, Security Hub and Detective are all enabled.

What addresses each need?

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|--|--|
| <p>A Extended Threat Detection correlates the findings; Detective answers what the role touched</p> | <p>B Detective correlates the findings; Security Hub answers what the role touched</p> |
| <p>C Security Hub correlates the findings; GuardDuty answers what the role touched</p> | <p>D A composite CloudWatch alarm correlates; CloudTrail Lake answers what the role touched</p> |

◆ PAUSE AND DECIDE

What addresses each need?

- A** Extended Threat Detection, then Detective
Attack sequences correlate signals into one critical finding, and Detective builds the behaviour graph for what a principal touched.
- B** Detective correlates
Detective investigates a finding that already exists. It does not produce the correlated finding in the first place.
- C** Security Hub correlates
Security Hub aggregates findings into one place, which is not the same as correlating them into one incident.
- D** Composite alarm and Lake
Composite alarms combine alarms, not findings. CloudTrail Lake is also in maintenance and not the investigation tool here.

THE SIGNAL

Three services, three verbs. GuardDuty detects, Security Hub aggregates, Detective explains. Aggregating is not correlating.

A company must move a self-managed MySQL database on EC2 to Amazon RDS for MySQL with under ten minutes of downtime. Separately, an Oracle database must become Aurora PostgreSQL, and the team has budgeted two weeks of schema work for each.

Both are 500 GB.

What should the solutions architect advise?

- ☐ A MySQL is homogeneous and needs no schema conversion; Oracle to Aurora PostgreSQL needs SCT and far more than two weeks for code objects
- ☐ B Both need schema conversion, and two weeks each is reasonable
- ☐ C Neither needs schema conversion, since DMS handles both
- ☐ D MySQL needs conversion because the host changes; Oracle does not, since DMS converts automatically

◆ PAUSE AND DECIDE

What should the solutions architect advise?

- A** Homogeneous needs none; heterogeneous needs much more

Same engine means the dialect is identical, so nothing converts. An engine change means code objects need rewriting, which is rarely two weeks.
- B** Both need it

MySQL to MySQL is a hosting change, not an engine change. Budgeting conversion work for it spends effort on nothing.
- C** Neither needs it

DMS moves data. It does not convert an Oracle schema into PostgreSQL, which is SCT's job.
- D** Inverted

Exactly backwards on both halves. Changing host is not changing engine, and DMS does not convert schemas.

THE SIGNAL

Self-managed to managed feels like a big change and is only a hosting change. The engine is what decides.

A disaster recovery design keeps a database replicating continuously to a second Region, with compute defined in launch templates but not running. Auto Scaling will launch instances during a Region failure. Separately, an application's configuration keeps drifting back after engineers edit it during incidents.

The RTO is two hours.

What describes each situation correctly?

- ☐ A That is a pilot light, and it depends on the control plane; the drift needs a State Manager association
- ☐ B That is warm standby, and it is statically stable; the drift needs Run Command
- ☐ C That is a pilot light, and it is statically stable; the drift needs Run Command
- ☐ D That is warm standby, and it depends on the control plane; the drift needs a State Manager association

◆ PAUSE AND DECIDE

What describes each situation correctly?

- A

 Pilot light, control plane dependent, State Manager
Nothing is serving, so it is pilot light. Launching instances at failure is a control plane action. And "keeps drifting" means continuous, which is State Manager.
- B

 Warm standby, statically stable, Run Command
Wrong on all three. Nothing is running, launching is control plane work, and Run Command applies once.
- C

 Pilot light, statically stable, Run Command
Right on pilot light. Statically stable would mean nothing needs creating, and this design creates instances at failure.
- D

 Warm standby, control plane, State Manager
Right on the last two. Warm standby requires a scaled-down copy actually serving, and nothing is running here.

THE SIGNAL

Three pairs in one question, and every option gets some right. Resolve each independently rather than picking the familiar shape.

What to carry out of this module

- ◆ **Name the pair before reading the options.** Most questions are one of about thirty distinctions.
- ◆ **Learn the divider, not two definitions.** One property decides each pair, and everything shared is noise.
- ◆ **Composite questions resolve independently.** Each pair eliminates two options, so two pairs leave one answer.
- ◆ If both members still seem right, you have the wrong pair. Re-read the requirement rather than the options.

ON THE EXAM

The five most-missed: association versus propagation, reserved versus provisioned, governance versus compliance, coverage versus utilisation, and Detective not detecting.